

Time Spent: 300

Subject:

Biology

QID: 404100

System:

Musculoskeletal System

Potential is a phenomenon in which skeletal muscle force production in response to the same depolarizing stimulus increases during a series of contractions. The activity of an enzyme called myosin light chain kinase (MLCK) is thought to play an important role in potentiation.

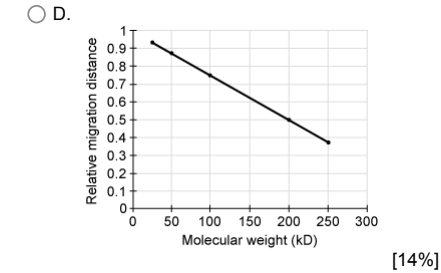
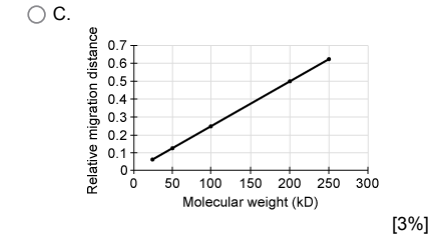
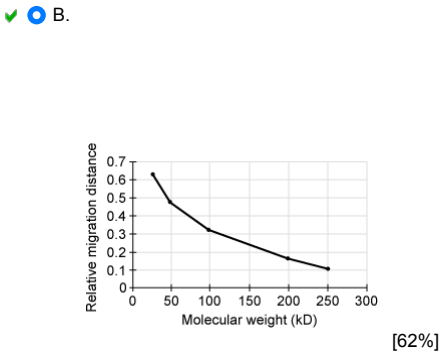
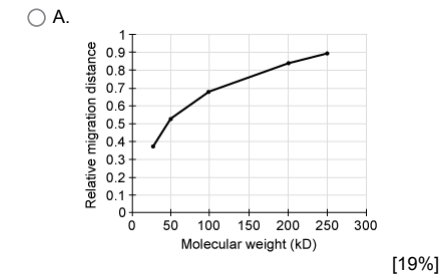
MLCK phosphorylates a protein called myosin regulatory light chain (RLC), which is a part of the myosin molecules composing the thick filament. The activity of MLCK is regulated by the binding of the protein calmodulin (CaM), which occurs when CaM binds Ca^{2+} released from the sarcoplasmic reticulum. Binding of the Ca^{2+} /CaM complex triggers a rearrangement of MLCK in which an autoinhibitory region moves away from the active site for RLC phosphorylation. RLC phosphorylation, which has been shown to vary with the frequency and strength of contractions, results in repositioning of the myosin heads slightly closer to actin binding sites, thereby facilitating actin–myosin interactions.

To investigate the role of myosin RLC phosphorylation in skeletal muscle function, researchers studied *in vitro* force production by the extensor digitorum longus (EDL), a fast-twitch skeletal muscle, of wild-type mice and mice with no skeletal muscle MLCK (skMLCK KO). The EDL muscles were electrically stimulated to produce several very brief ("twitch") submaximal contractions, each separated by a minute of rest. These baseline contractions were followed by a much stronger "sensitizing" contraction followed several seconds later by another twitch contraction. Within each set of mice, the forces of the baseline twitch contractions were similar. However, the final twitch contraction was significantly elevated in wild-type but not skMLCK KO EDL muscles. The average force produced by the twitch contractions before and after the sensitizing contraction are shown in Table 1 and are expressed relative to the average baseline force in wild-type muscles.

Table 1 Force Produced During Twitch Contractions Before and After the Sensitizing Contraction

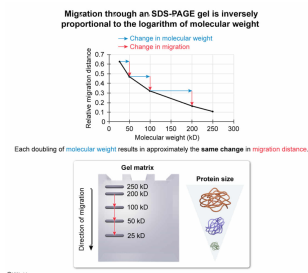
Group	Condition (before or after sensitizing contraction)	Relative force
Wild-type	Before	1.0
	After	2.1
skMLCK KO	Before	0.9
	After	1.1

Assume that the researchers measured skeletal muscle MLCK expression using SDS-PAGE. Proteins of known molecular weights were run in one lane as markers showing the logarithmic relationship between relative mobility on the gel and molecular weight. Which of the following shows the likely relative positions of molecular weight markers run by the researchers as they measured MLCK expression?



Correct
62% answered correctly

Explanation:



Electrophoresis is one means of separating proteins based on size and other properties (eg, charge). **Nondenaturing gel electrophoresis** maintains the size, charge, and shape of the proteins being separated, whereas **denaturing gel electrophoresis** disrupts **protein structure** and controls the charge on the proteins.

The **crosslinking** of electrophoresis gels forms spaces that are sometimes called "pores." In both nondenaturing and denaturing gel electrophoresis, these pores influence the passage of proteins through the gel during electrophoresis, with larger proteins generally more impeded (ie, slower moving) than smaller proteins.

In a type of electrophoresis known as **SDS-PAGE**, molecules of sodium dodecyl sulfate (SDS) coat and denature proteins. The SDS also imparts a relatively uniform amount of negative charge relative to the size of the proteins (ie, the mass-to-charge ratio of the proteins is approximately the same). As a result, SDS-PAGE separates proteins primarily based on size.

The passage describes an experiment in which a team of researchers studied mice that either did or did not have MLCK in their skeletal muscles. The question asks for the graph showing the likely relative positions of molecular weight markers run by the researchers as they measured MLCK expression using SDS-PAGE.

The **relative mobility** of a protein separated using SDS-PAGE is **inversely proportional** to the protein's **molecular weight**, and, as stated in the question, the **relationship between mobility and molecular weight** is **logarithmic**. The only answer choice graph showing a **logarithmic, inversely proportional relationship** between migration distance and molecular weight is **Choice B**.

(Choices A and C) The relative migration is negatively (*not* positively) correlated with molecular weight.

(Choice D) The relationship between a number and its logarithm is curved (*not* linear).

Educational objective:

SDS-PAGE separates denatured proteins primarily based on size.

Time spent: 0 sec
Subject:
Biology

QID: 404186
System:
Musculoskeletal System

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MLCK phosphorylates a protein called myosin regulatory light chain (RLC), which is a part of the myosin molecules composing the thick filament. The activity of MLCK is regulated by the binding of the protein calmodulin (CaM), which occurs when CaM binds Ca²⁺ released from the sarcoplasmic reticulum. Binding of the Ca²⁺/CaM complex triggers a rearrangement of MLCK in which an autoinhibitory region moves away from the active site for RLC phosphorylation. RLC phosphorylation, which has been shown to vary with the frequency and strength of contractions, results in repositioning of the myosin heads slightly closer to actin binding sites, thereby facilitating actin–myosin interactions.

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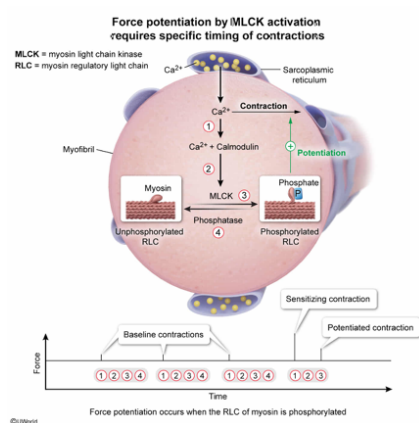
Group	Condition (before or after sensitizing contraction)	Relative force
Wild-type	Before	1.0
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skMLCK KO	Before	0.9
	After	1.1

Which of the following most likely explains why force production in wild-type muscles was NOT increased above baseline until after the sensitizing contraction was completed during the series of contractions carried out in the study?

- ☐
A. The rate of ATP synthesis was not sufficient for phosphorylation of the RLC before the first contraction. [21%]
- ☐
B. The hydrolysis of ATP is a slow process. [4%]
- ☒
C. The phosphorylation state of RLC varies with time and contraction intensity. [54%]
- ☐
D. The increased amount of Ca²⁺/CaM complex formed at low cytosolic Ca²⁺ concentration inhibits potential. [19%]

Correct
54% answered correctly

Explanation:



Skeletal muscle force production is dependent on the activity of the cross-bridge cycle, in which **myosin** interacts with **actin** in a Ca²⁺-dependent process that engages other thin filament proteins (eg, troponin and tropomyosin). The force-generating step in this process occurs when **myosin pulls** on actin.

The passage describes modulation of force production due to the **phosphorylation** of myosin by MLCK, noting that the forces produced by the mouse EDL muscles were similar during baseline contractions of both wild-type and skMLCK KO muscles. The question asks why force production was not increased above baseline until after the sensitizing contraction in the wild-type muscles.

Before muscle force can be potentiated by MLCK, Ca^{2+} must first bind CaM, then the Ca^{2+} /CaM complex must bind MLCK, and MLCK must subsequently phosphorylate the RLC of myosin. In addition, the RLC must remain phosphorylated until the next contraction occurs. Because RLC phosphorylation is dependent upon contraction intensity, the sensitizing contraction results in a **greater degree of RLC phosphorylation**. Initiating another twitch contraction soon after the sensitizing contraction ensures that RLC remains phosphorylated. Therefore, the **likely reason** that **force was not increased above baseline** until **after the sensitizing contraction** is that **myosin RLC phosphorylation** depends on the **timing and intensity of contractions**.

(Choice A) ATP synthesis *would* have increased after the first contraction to replace the ATP that was used (ie, for the cross-bridge cycle and for transporting Ca^{2+} back into the sarcoplasmic reticulum). However, there is no information given to suggest that ATP supply for RLC phosphorylation was limited at baseline.

(Choice B) ATP hydrolysis is fast enough to enable the power stroke to be completed during multiple cross-bridge cycles during a sustained contraction and, within the context of the experiment described, during multiple repetitive contractions. Therefore, the delayed increase in force is *not* due to a slow rate of ATP hydrolysis.

(Choice D) Ca^{2+} /CaM complex formation would be decreased (*not* increased) at a low cytosolic Ca^{2+} concentration.

Educational objective:

The force-generating step in muscle contraction occurs when myosin pulls on actin, and myosin phosphorylation by MLCK is a means by which this process can be modulated.

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Biology

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Musculoskeletal System

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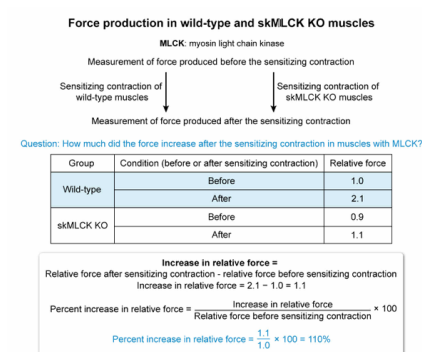
In muscles with MLCK, the sensitizing contraction increased the relative force of the twitch contraction by:

- ☐ A. 11%. [7%]
- ☐ B. 22%. [14%]
- ☐ C. 91%. [4%]
- ☒ D. 110%. [73%]

Correct

72% answered correctly

Explanation:



Muscles produce force through the interaction between actin and myosin molecules in sarcomeres. The **force-generating step** in the cross-bridge cycle (ie, pulling on actin by the myosin head) is Ca^{2+} dependent and can be modulated by other processes (eg, **phosphorylation and dephosphorylation**) in the muscle fiber, including the Ca^{2+} -dependent activation of the enzyme MLCK.

Table 1 shows the force produced by wild-type (WT) and skMLCK KO EDL muscles, and the question asks for the increase in relative force that occurred after the sensitizing contraction in muscles with MLCK. Only WT muscles contain MLCK, so the increase in force can be calculated as follows:

Increase in relative force = Relative force after sensitizing contraction – Relative force before sensitizing contraction

$$\text{Increase in relative force} = 2.1 - 1.0 = 1.1$$

$$\text{Percent increase in relative force} = \frac{\text{Increase in relative force}}{\text{Relative force before sensitizing contraction}} \times 100$$

$$\text{Percent increase in relative force} = \frac{1.1}{1.0} \times 100 = 110\%$$

Therefore, the **increase in relative force** that was **induced by the sensitizing contraction** in muscles **containing MLCK** (ie, WT muscles) **was 110%**.

(Choices A, B, and C) These choices reflect incorrect consideration of the percent increase in the relative force of the twitch contraction that occurred after the sensitizing contraction.

Educational objective:

The force-generating step in the cross-bridge cycle occurs when the myosin head pulls on actin, and this process can be affected by processes such as MLCK phosphorylation of the myosin RLC.

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